Cannula Filtration/Transfer

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Category: Protocol
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Goal

Filtration or transfer of a solution under argon atmosphere.

When should I use a syringe, frit technique or canula technique?

Syringes are used only for the transfer of solvents.

Frit filter technique shoud be used to filter a suspension with the aim to gain the solid. The suspension is transfered froom one vessel into the other (with a cannula) through creating static vaccum over the tube of the schlenk line.

Cannula (PTFE tube) have a high inside diameter which allows the transfer of suspensions with small particles. Cannulas equipped with a filter can be used to filter a suspension and to gain a homogeneous solution. The suspension is transferred from one flask into the other by pushing it with high pressure of argon.

Prerequisites and preparation

Cannula ("white" PTFE tube), Filter, Tape

2x Septum

dummy flask for changing the atmosphere of the cannula to argon

Schlenk flask

solvent, syringe for washing

Steps for filtration

Step number	Step description
1. Build the cannula system	The cannula is punctured on both sides into a septum. One side is placed inside an empty Schlenk.
	The other is side is made even with a straight cut. The cannula is wickled by tape, filter and again tape. Finally, it is placed with the septum inside a dummy flask
2. Change the atmosphere to Ar	The atmosphere in the Schlenk and in the cannula is changed by evacuating and subsequently refilling with argon three times.
3. Connect the cannula with the reaction schlenk	While beeing under argon atmosphere the cannula side with the filter is placed in the reaction flask.

4. Pressure balance with metal cannula	The septum of the empty Schlenk is punctured with a small (blue) metal cannula. The stop cock can be closed a bit, but ALWAYS a small argon flow should be sensed/heard outgoing from the metal cannula.
5. Filtration/Transfer	By closing (not completely) cautionley the stop cock on the schlenk, the argon flow is reduced causing a pressure difference between this schlenk and the schlenk with the suspension. Thus, the solution is pressed through the filter inside the cannula towards the empty Schlenk. The velocitiy of the transfer can be adjust with opening and closing of the stop cock (variing the pressure difference between the flasks)
6. Washing	The residue of the suspension can easily be washed with solvent by putting a syringe inside the septum. Stepwise the solvent can than be added without removing out the syringe. CAREFULL, do not open the solvent while the metall canula is stucked in the solution, because the opening of another schlenkline to argon reduce the argon gasflow.
7. Close	After washing the argon stop cock should be open again completely and the metal cannula as well as the septum and the PTFE cannula can be removed. THe Schlenk should be closed with a plug.

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